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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

AM4333-1 – PROMPT ENGINEERING
(For AM)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What model type did Transformers replace in NLP tasks?**
A) CNN
B) LSTM
C) SVM
D) Decision Trees
2. **What is a key characteristic of Transformer models?**
A) Use of recurrence
B) Use of self-attention mechanism
C) Fixed sequence processing
D) Requires labeled data only
3. **Which component of a Transformer model enables parallel processing?**
A) Recurrent layers
B) Attention mechanism
C) Dropout layers
D) Pooling layers
4. **What makes Transformers suitable for parallel computation?**
A) Independence of input tokens
B) Recurrent processing
C) Use of activation functions
D) Fixed sequence dependency
5. **What is the main application of Transformer models?**
A) Image classification
B) Time series forecasting
C) Natural language processing
D) Anomaly detection
6. **What is an advantage of Transformers over traditional RNNs?**
A) They require less data
B) They use fixed word embeddings
C) They allow parallel processing
D) They do not require training
7. **What is prompt engineering primarily used for?**
A) Enhancing model efficiency
B) Optimizing responses from LLMs
C) Improving database performance
D) Reducing training time
8. **Zero-shot prompting refers to:**
A) Providing no examples in the prompt
B) Providing one example in the prompt
C) Using only labeled data
D) Training from scratch
9. **Chain-of-thought prompting is useful for:**
A) Improving model interpretability
B) Reducing processing time
C) Removing biases
D) Handling adversarial attacks
10. **Which prompting method is best for sentiment analysis?**
A) Zero-shot prompting
B) Chain-of-thought prompting
C) Few-shot prompting
D) Graph prompting
11. **Which of the following is an example of a domain-specific LLM?**
A) GPT-4
B) BioBERT
C) DALL·E
D) LaMDA
12. **Which setting in an LLM helps control randomness in responses?**
A) Context window
B) Temperature
C) Token limit
D) Learning rate

13. What is one way to improve prompt effectiveness?
 A) Using clear and specific instructions B) Increasing model parameters
 C) Reducing dataset diversity D) Lowering temperature settings
14. What makes a good prompt for factual accuracy?
 A) Including source references B) Using minimal context
 C) Randomized responses D) Reducing token limit
15. What is Hugging Face primarily known for?
 A) Cloud computing B) NLP model hosting and development
 C) Social media analytics D) Image recognition
16. What does the Transformers library primarily offer?
 A) Cloud storage B) NLP models and training utilities
 C) Image recognition APIs D) Social media trend analysis
17. What type of tasks does the Hugging Face API simplify?
 A) Complex database queries B) Machine learning inference
 C) Web development D) Cybersecurity measures
18. How does Hugging Face's Datasets library help in machine learning workflows?
 A) Provides a scalable way to load, preprocess, and share datasets B) Eliminates the need for labeled training data
 C) Focuses solely on synthetic dataset generation D) Restricts dataset access to premium users
19. What type of AI applications can be deployed using Hugging Face's Spaces?
 A) Interactive chatbots B) Cryptocurrency wallets
 C) Cloud security solutions D) Network simulations
20. What is a key advantage of Hugging Face Datasets?
 A) Easy integration with ML workflows B) Automatic dataset labeling
 C) Generating synthetic data D) Image recognition support

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain how transformer models differ from RNN-based models like LSTMs.	8	L*2	1	1
	b) Illustrate the encoder workflow in transformer architecture with a diagram.	8	L3	1	1
2.	a) Apply the concept of pre-training and fine-tuning in LLMs with examples.	8	L3	2	2
	b) Apply the concept of masked language modeling in BERT with an example.	4	L3	2	1
	c) Differentiate between general purpose and domain specific LLMS with examples.	4	L3	2	1
3.	a) Describe the key components of the decoder in transformer models.	8	L2	2	2
	b) Explain the role of self-attention in transformer models.	8	L2	2	1
Unit – II					
4.	a) Identify the key elements of an effective prompt and their roles.	8	L1	3	2
	b) Apply zero-shot, few-shot, and chain-of-thought prompting techniques to a simple task.	8	L3	3	1
5.	a) Demonstrate Adversarial prompting and factuality in risk and misuses.	8	L3	3	2
	b) Explain any 5 "Example of prompts"	8	L3	4	2

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| 6. a) | Apply a structured prompting approach to improve sentiment analysis accuracy. | 8 | L3 | 4 | 1 |
| b) | Interpret how biases in prompts can impact model outputs. | 8 | L3 | 4 | 2 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Apply Gradio to create a simple interactive NLP application. | 8 | L3 | 5 | 1 |
| b) | Summarize the key features of the Hugging Face platform. | 4 | L2 | 5 | 2 |
| c) | Illustrate the components of the Hugging Face library with examples. | 4 | L3 | 5 | 2 |
| 8. a) | Analyze the advantages and challenges of using cloud-based NLP tools. | 8 | L3 | 5 | 1 |
| b) | Summarize the process of using Hugging Face datasets for model training. | 8 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
